

Myocardial Positron Emission Tomography (PET) and PET-CT

ACG: A-0097 (AC)

[Link to Codes](#)

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Clinical Indications

- Myocardial positron emission tomography (PET), with or without simultaneous computed tomography (PET-CT), may be indicated for **1 or more** of the following(1)(2)(3)(4):
 - ☐ Cardiac implantable electronic device (eg, pacemaker, implantable cardioverter-defibrillator, cardiac resynchronization therapy device) infection, suspected, as indicated by **ALL** of the following(10)(11)(12)(13)(14)(15)(16)(17):[N](#)
 - Clinical suspicion of device infection
 - Echocardiogram results are nondiagnostic.
 - Study performed using PET-CT
 - ☐ Coronary artery disease assessment,^[A] as indicated by **1 or more** of the following(21)(22)(23)(24)(25):[N](#)
 - Asymptomatic patient with elevated troponin level(54)
 - Asymptomatic patient with ischemic changes present on resting ECG
 - Cardiac transplant allograft vasculopathy surveillance
 - Congenital heart disease, known (eg, repaired aortic coarctation, previous coronary reimplantation surgery, coronary artery compression)(53)
 - Exercise treadmill test results equivocal or positive, requiring further evaluation
 - Need for ischemic evaluation, as indicated by **ALL** of the following:
 - Clinically significant coronary artery disease, as indicated by **1 or more** of the following(26):
 - Anomalous coronary artery, known, and need for functional assessment(52)
 - Before revascularization (ie, percutaneous coronary intervention or coronary artery bypass grafting) to demonstrate ischemia
 - Cardiac CT angiography with 50% or greater stenosis of at least one coronary artery, or inconclusive results
 - Coronary calcium Agatston score 1 to 99 on cardiac CT scan, and high risk for coronary heart disease, as indicated by **1 or more** of the following(23)(55)(56)(57)(58):
 - Calculated 10-year absolute coronary heart disease risk 20% or greater by global risk calculator(59)(60)
 - Coronary heart disease risk equivalent (eg, peripheral arterial disease, carotid artery disease, or abdominal aortic aneurysm)
 - Diabetes mellitus in patient age 40 years or older

- Prior stroke or transient ischemic attack(61)
 - Coronary calcium Agatston score of 100 or more on cardiac CT scan(55)(56)(57)
 - Detection of recurrent stenosis or progression of disease in asymptomatic patient who had silent ischemia prior to revascularization and **1 or more** of the following(21)(23)(62):
 - Diabetes
 - Five years or more after coronary artery bypass grafting
 - History of prior cardiac arrest
 - Incomplete revascularization procedure (ie, persistent stenosis after percutaneous coronary intervention or coronary artery bypass grafting)
 - Involvement of proximal left anterior descending artery
 - Left ventricular ejection fraction of 35% or less
 - Two years or more after percutaneous coronary intervention (ie, angioplasty or stent)
 - Following acute coronary syndrome and **ALL** of the following(21)(63):
 - Absence of current ischemic symptoms
 - No signs of heart failure or hemodynamic instability(64)
 - Not planning to undergo early revascularization
 - Following ST-segment elevation myocardial infarction
 - History of Kawasaki disease with known coronary artery aneurysm, to demonstrate inducible ischemia(65)
 - Increase in anginal frequency or severity(64)
 - New heart failure symptoms or increase in existing heart failure symptoms(66)(67)(68)
 - New onset of ischemic symptoms without ECG changes(64)
 - New ST-segment changes on ECG at rest
 - Ventricular arrhythmia
 - Exercise treadmill testing alone is or would be unreliable or inconclusive due to **1 or more** of the following(21)(69)(70)(71):
 - Anginal symptoms induced by exercise in high-risk patient without any diagnostic ECG changes on previous exercise treadmill test(52)
 - Atrial fibrillation or flutter
 - Electronically paced ventricular rhythm
 - Insufficient workload (ie, inability to achieve at least 85% of maximum age-sex predicted heart rate on Bruce protocol exercise treadmill testing)(1)
 - Left bundle branch block(72)
 - Left ventricular hypertrophy with repolarization abnormalities on baseline ECG
 - Patient taking digoxin
 - Pre-excitation syndrome (eg, Wolff-Parkinson-White syndrome)
 - ST segment on baseline ECG with 1 mm depression or greater
- ☐ Endocarditis, known or suspected, as indicated by **ALL** of the following(12)(13)(17)(73)(74)(75)(76)(77):[\[B\]](#)
- Clinical scenario as indicated by **1 or more** of the following:
 - Clinical suspicion of endocarditis (eg, persistent bacteremia, new or worsened heart murmur)
 - Perivalvular complication of endocarditis suspected (eg, abscess)
 - Echocardiogram results are nondiagnostic or equivocal.
 - Study performed using PET-CT
- ☐ Myocardial viability assessment,[\[B\]](#) as indicated by **1 or more** of the following(6)(23)(24)(46):[\[B\]](#)
- Chronic secondary mitral regurgitation, and patient is candidate for revascularization if viable myocardium identified(74)(90)
 - Ischemic cardiomyopathy, known or suspected, with left ventricular ejection fraction 35% or less(5)
 - Preoperative or preprocedural planning needed for revascularization (ie, coronary artery bypass graft or percutaneous coronary intervention), if viable myocardium identified(86)(89)
 - SPECT myocardial perfusion scan or stress echocardiogram findings inconclusive, or no viable myocardium evident(91)
- Sarcoidosis with suspected cardiac involvement(67)(68)(77)(92)[\[B\]](#)
- ☐ Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
- Clinical need for repeat imaging, as indicated by **1 or more** of the following:
 - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
 - Need for interval reassessment that may impact treatment plan
 - Need for re-imaging either prior to or after performance of invasive procedure
 - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

Evidence Summary

Background

Myocardial PET is a noninvasive nuclear medicine technique that uses positron-emitting radionuclides, such as F-18 fluorodeoxyglucose, rubidium-82, or ammonia N-13, to obtain images of myocardial viability, perfusion, and metabolism.(4)(5)(6) **(EG 2)** Images are acquired at rest and with stress (either exercise or pharmacologic), and a defect in the image with stress that is not present at rest usually indicates an area of myocardial ischemia.(5)(7) **(EG 2)** Myocardial PET with ECG gating can assess ventricular function, including ejection fraction and segmental contractility, in addition to myocardial perfusion.(5) **(EG 2)** PET imaging involves exposure to ionizing radiation, and research is under way to develop scanning techniques that can reliably reduce the radiation dose delivered in cardiac PET imaging.(1)(8)(9) **(EG 2)** Hybrid scanners integrate diagnostic CT scanners into PET scanners in order to provide both anatomic and functional information in a single study.(4) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 74 published peer reviewed articles, 25 specialty society or other evidence-based guidelines, and 1 book section.

For cardiac implantable electronic device infection, suspected, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A meta-analysis of 14 observational studies (492 patients) found that F-18 fluorodeoxyglucose PET-CT had a pooled sensitivity and specificity of 85% and 90%, respectively, when used to diagnose cardiac implantable electronic device infections.(18) **(EG 1)** An expert consensus statement endorsed by multiple specialty societies recommends F-18 fluorodeoxyglucose PET-CT imaging as an additive diagnostic test for suspected cardiac implantable electronic device infections with nondiagnostic echocardiogram or unknown extent of infection.(10) **(EG 2)**

For coronary artery disease assessment, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Myocardial perfusion imaging is useful for detecting the presence of new or worsening ischemia suggestive of myocardium-at-risk and identifying the need for changes to treatment plans, including revascularization; it is also used to detect restenosis or new areas of stenosis following revascularization.(21)(26)(27)(28) **(EG 2)** There is increasing evidence for the prognostic value of myocardial PET in perfusion imaging; where available, myocardial PET may be used to evaluate myocardial perfusion and function in patients with known or suspected coronary artery disease.(29)(30)(31)(32) **(EG 2)** Review articles note that the high spatial resolution of myocardial PET, combined with its ability to calculate absolute coronary flow reserve based on quantitative measures that have been directly correlated with ischemia, makes it an ideal imaging modality for assessing myocardial perfusion.(1)(33)(34) **(EG 2)** Average sensitivity of PET perfusion imaging for detecting angiographic coronary stenosis of greater than 50% has been reported to be 91%, with average specificity of 89%.(35) **(EG 2)** A multicenter study of 372 patients with suspected obstructive coronary artery disease on cardiac CT angiography (a diameter exceeding 2 mm and 50% to 100% reduction of diameter) found that rubidium-82 PET myocardial perfusion imaging had a sensitivity and specificity of 64% and 89%, respectively, for identifying hemodynamically significant obstructive coronary artery disease (as compared with the reference standard of invasive coronary angiography with fractional flow reserve).(36) **(EG 2)** Myocardial PET has been demonstrated to have improved image quality, less attenuation, reduced scanning time, and greater diagnostic accuracy than myocardial SPECT.(31)(37)(38)(39) **(EG 2)** Myocardial PET perfusion imaging has also been noted to deliver a lower average effective radiation dose than SPECT perfusion imaging.(1)(40) **(EG 2)** A meta-analysis of 166 studies found that, among the cardiac imaging modalities of SPECT, cardiac MRI, and myocardial PET, myocardial PET achieved the highest diagnostic performance, with a pooled sensitivity of 84% and pooled specificity of 81%.(41) **(EG 2)** A retrospective cohort study of 14,649 patients undergoing PET or PET-CT myocardial perfusion imaging found, at a median follow-up of 3.4 years, that left ventricular ejection fraction reserve (the difference in ejection fraction between stress and rest periods) of 0 or less was associated with lower rates of cardiac death with early revascularization, compared with medical therapy (even after adjusting for severity of ischemia), in patients without known coronary artery disease at baseline; patients with left ventricular ejection fraction reserve values of greater than 0 had similar survival with revascularization and medical therapy.(42) **(EG 2)** Analysis of a multicenter observational registry that included 7061 patients who underwent a rest/stress rubidium-82 PET myocardial perfusion scan concluded that, in patients with known or suspected coronary artery disease, PET myocardial perfusion imaging provided incremental cardiac risk estimates as compared with traditional coronary risk factors, regardless of BMI.(43)(44) **(EG 2)** Retrospective analysis of a prospective observational registry including 6037 patients with suspected cardiac symptoms concluded that stress rubidium-82 PET provided clinically meaningful risk stratification of both women and men, and was particularly helpful for identifying high-risk older women (age 55 years and older).(45) **(EG 2)** A consensus statement endorsed by multiple specialty societies states that myocardial PET, particularly stress myocardial perfusion PET, is appropriate for evaluation of heart failure suspected to be ischemic in etiology.(46) **(EG 2)** A prospective study of 140 patients who underwent heart transplant found that abnormalities on rubidium-82 myocardial PET imaging were predictive of adverse cardiac events due to allograft vasculopathy, as well as all-cause mortality; the authors caution that larger prospective trials are needed to confirm these findings.(47) **(EG 2)** Review articles note that coronary flow reserve measurements from myocardial PET-CT can be useful in patients presenting with ischemia with nonobstructive coronary arteries, helping to identify high-risk patients and guide management.(48)(49) **(EG 2)** Review articles note that although invasive coronary angiography is the gold standard diagnostic test, cardiac PET imaging is useful for noninvasive surveillance for cardiac allograft vasculopathy in patients with a history of cardiac transplant.(20)(50) (51) **(EG 2)** Review articles note that myocardial perfusion stress imaging is appropriate to detect ischemia and evaluate function in patients with an anomalous aortic origin of a coronary artery.(52)(53) **(EG 2)** A review article notes that certain congenital heart conditions are associated with risk factors for ischemic heart disease (eg, persistent hypertension in patients with repaired aortic coarctation); coronary artery compression syndromes and coronary fistulas can cause coronary ischemia, as can congenital heart surgeries that involve coronary reimplantation. Myocardial perfusion stress imaging, including myocardial PET, can detect ischemic heart disease in these patients.(53) **(EG 2)**

For endocarditis, known or suspected, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Initial echocardiography has a high sensitivity and specificity for native valve endocarditis but has been found to be normal or inconclusive in nearly 30% of cases of prosthetic valve endocarditis.(78)(79) **(EG 2)** A systematic review evaluating the diagnostic performance of imaging modalities in infective endocarditis included 13 studies of PET-CT with F-18 fluorodeoxyglucose and found, for the diagnosis of prosthetic valve endocarditis, sensitivities of 73% to 100% and specificities of 71% to 100%; when added to modified Duke criteria, PET-CT increased the sensitivity of the criteria from a range of 52% to 70% to a range of 91% to 97%.(80) **(EG 1)** Specialty society guidelines have included positive findings on molecular nuclear imaging techniques, such as myocardial PET-CT, as major criteria for the diagnosis of cardiac valve infections.(15)(73)(74)(81) **(EG 2)** Specialty society guidelines state that PET-CT with F-18 fluorodeoxyglucose may be an appropriate additional imaging test to help guide treatment in suspected infective endocarditis.(13)(82) **(EG 2)**

For myocardial viability assessment, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Myocardial PET scan with F-18 fluorodeoxyglucose is considered the gold standard for the assessment of myocardial perfusion and viability(5)(83)(84) because of its correlation with improvement in left ventricular function following revascularization.(85)(86)(87) **(EG 1)** Myocardial PET scan with F-18 fluorodeoxyglucose may help with selection of candidates for revascularization in patients with extensive coronary disease and left ventricular dysfunction.(66)(85)(86) **(EG 1)** Myocardial PET scan with F-18 fluorodeoxyglucose has been shown to alter clinical decision making for at least 60% of patients who have chronic left ventricular dysfunction.(88) **(EG 2)** A specialty society guideline states that assessment of myocardial viability is reasonable prior to revascularization in heart failure patients with coronary artery disease.(89) **(EG 2)** Appropriate-use criteria endorsed by multiple specialty societies state that myocardial PET is appropriate to assess for myocardial viability in chronic secondary mitral regurgitation, which can be improved with revascularization if due to an ischemic etiology and viable myocardium is found.(90) **(EG 2)**

For sarcoidosis with suspected cardiac involvement, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Myocardial PET imaging with F-18 fluorodeoxyglucose has enhanced detection of cardiac sarcoidosis, and imaging with myocardial PET can be useful both for identification of cardiac sarcoidosis, risk stratification, and monitoring response to treatment.(77)(93)(94)(95)(96) **(EG 2)** A retrospective study of 107 patients referred for evaluation of cardiac sarcoidosis compared the diagnostic utility of cardiac MRI and myocardial PET scan; when PET data were added to cardiac MRI findings, 45% of patients were reclassified as having a higher or lower likelihood of cardiac sarcoidosis, and 80% of these reclassifications were ultimately correct. The authors stated that these 2 imaging modalities provided complementary value in diagnosing cardiac sarcoidosis and guiding patient management.(97) **(EG 2)** A review article notes that myocardial F-18 fluorodeoxyglucose PET may be used to evaluate myocarditis associated with cardiac sarcoidosis.(98) **(EG 2)** Specialty society guidelines state that myocardial PET scanning may be used to identify cardiac involvement in sarcoidosis.(99)(100)(101)(102) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(103); 42 CFR 422.101(104)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(105); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(106); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(107)

Medicare Coverage Determinations: Medicare Coverage Database(108)

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Footnotes

[A] The use of PET imaging for myocardial ischemia (ie, myocardial perfusion imaging) is performed using rubidium-82 or ammonia N-13.(1)(4)(19)(20) [A in Context Link 1, 2]

[B] Imaging for myocardial viability by PET is performed using F-18 fluorodeoxyglucose.(1)(4)(6)(19) [B in Context Link 1, 2]

Codes

CPT®: 78429, 78430, 78431, 78432, 78433, 78434, 78459, 78491, 78492, 78814

HCPCS: A9526, A9552, A9555, A9598, A9609, S8085

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